9 Factor the following:

a $b^2 + 14b + 49$

b $y^2 - 10y + 25$

c $x^2 + 10x + 25$

d $x^2 + 14x + 49$

e $x^2 - 12x + 36$

f $x^2 - 16x + 64$

g $81x^2 - 72x + 16$

h $81x^2 + 36x + 4$

i $49x^2 - 28x + 4$

j $81 + 18x + x^2$

k $36 - 12x + x^2$

l $25 + 60r + 36r^2$

m $7d^2 + 112d + 448$

n $4x^2 + 40x + 100$

o $3x^2 - 24xy + 48y^2$

p $-6r^2 - 96r - 384$

q $-3x^2 + 12x - 12$

r $b^2 - 7b + \frac{49}{4}$

s $x^2 + 3x + \frac{9}{4}$

t $x^2 - 3x + \frac{9}{4}$

10 Factor the quadratics using the method of completing the square to get them into the form:

$$y = \left(x - a + \sqrt{b}\right)\left(x - a - \sqrt{b}\right)$$

a $y = x^2 - 6x + 7$

b $y = x^2 + 8x + 14$

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